MILITARY SPECIFICATION

CHARGER, BATTERY PP-1451( )/G

1. SCOPE

1.1 Scope. - This specification covers one type of rectifier-type, self-regulated battery charger, designated as Charger, Battery, PP-1451( )/G. (See 6.3)

2. APPLICABLE DOCUMENTS

2.1 Documents. - The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant, (For Carton Sealing)

* MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-131	Barrier Material, Water Vaporproof, Flexible
MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment

FSC 6130

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MIL-A-5092	Adhesive Rubber Base, General Purpose
MIL-P-11268	Parts, Materials, and Processes used in Electronic Communication Equipment
MIL-G-12803	Gasket Material, Non-Metallic
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment

* STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definition Of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electrical Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

* DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SC-DL-379595	Charger, Battery, PP-1451( )/G
SC-A-678002	Burn-In/Debugging Procedure for Charger, Battery PP-1451( )/G

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

* 3. REQUIREMENTS

3.1 Description. - Charger, Battery PP-1451( )/G is a full-wave silicon rectifier type power supply incorporating a pair of matched self saturating magnetic control toroids with feedback control for automatic voltage regulation. The equipment is designed to operate from either 115 volts ac, single phase, or 230 volts ac, single phase, at 50, 60, or 400 hertz per second. The nominal output is either 50 amperes at 27.5 volts dc, or 25 amperes at 55 volts dc.

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The overall dimensions are 22 inches wide by 19 1/2 inches deep by 21 inches high without shockmounts and 22 inches wide by 22 inches deep by 23 inches high with shockmounts. The weight is 285 pounds including shockmounts.

* 3.2 Preproduction samples. - The contractor shall furnish three (3) preproduction samples of Charger, Battery, PP-1451( )/G, for approval if required by the invitation for bids and contract. (See 4.3 and 6.2(d)(1)).

3.3 Construction. - The equipment shall be constructed in accordance with the requirements of the specification, and of Drawing SC-DL-379595 (including the requirements for parts, materials, and processes thereon).

* 3.4 Parts, materials, and processes. - In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 6.8).

3.5 Finish, protective. - Equipment shall be given protective finish in accordance with MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on type I surfaces shall be green color (olive drab), semi-gloss enamel matching a color chip provided by the procuring agency. (See 6.4)

3.6 Marking. - Marking shall conform to MIL-M-13231. (See 4.4) Front panel markings shall be group I as described in that specification.

3.6.1 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with mininum disassembly of the equipment.

3.6.2 Serial number. - Each Charger, Battery, PP-1451( )/G shall be serial numbered.

3.7 Adjustment and repair. - The equipment shall be so constructed that parts, terminals, wiring, etc., are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring and with use of the minimum number and variety of special tools, particularly those needed for tuning and adjustment. When tools for tuning or adjustment are needed, the unit where they are used shall provide convenient means for mounting them. This equipment has been designed so that the toroids used are matched in characteristic, however, a slight unbalance between them or other circuit component variations may be adjusted by means of potentiometers located on the input adjustment panel.

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3.8 Input voltage. - The battery charger shall be capable of operation (See 3.9) from the following single phase sources:

Volts AC (± 10%)	Frequency (HZ (± 5%)
115, 230	50, 60 and 400

3.9 Output. - The equipment shall fulfill the following output conditions with any combination of input conditions. (See 4.7)

3.9.1 Voltage. - The output voltage with the front panel control switch in the CHARGE position shall be variable, per 3.9.1.3, from 25 volts dc to 30 volts dc suitable for modified constant potential charging of lead acid or nickel cadmium batteries. The output voltage with the front panel control switch in the NORMAL position shall be variable, per 3.9.1.3, from 25 volts dc to 30 volts or from 50 volts dc to 60 volts dc depending on the range selection on the connection board within the equipment. (See 4.7.2)

3.9.1.1 Voltage regulation. - The voltage regulation from no load to full load on either of the output ranges shall be within ± 1.5 percent except at minus 40°F the regulation shall be within ± 2 percent after 5 minutes of warm-up operation. (See 4.7.3)

3.9.1.2 Ripple voltage. - The ripple voltage from no load to full load on either of the output ranges shall not exceed 1.5 percent rms of the dc voltage except at minus 40°F the ripple voltage shall not exceed 5 percent after 5 minutes of warm-up operation. (See 4.7.4)

* 3.9.1.3 Voltage control. - The voltage control rheostat on the front panel shall provide adjustment within the specified ranges and shall require over 200 degrees rotation to cover the range of either voltage.

3.9.2 Output current. - Charger, Battery,PP-1451( )/G shall be capable of providing the output current as specified in paragraphs 3.9.2.1 and 3.9.2.2. (See 4.7.5)

3.9.2.1 Normal position. - With the selector switch in the NORMAL position, the equipment shall provide a full load continuous duty current of 50 amperes at 27.5 volts and 25 amperes at 55 volts. Operation of 0 amperes to 25 amperes or 50 amperes, depending on the range, shall meet the requirements of 3.9.1 independent of the type of load applied (dynamotor, resistive, inductive or battery).

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* 3.9.2.2 Charge position. - With the selector switch in the CHARGE position, the maximum current drain from the charger shall be automatically limited to 80 amperes to prevent damage to the charger. The voltage shall begin dropping as the current increases above 20 amperes.

3.9.3 Overload and open circuit protection. - The equipment shall be protected from damage by any open circuit or overload condition (including a direct short circuit) in both the NORMAL and CHARGE operating positions. Output contactors shall be provided to automatically open the circuit when the charger is not in operation, or when the input circuit is accidently "open circuited" with the charger in operation. (See 4.7.6)

3.10 Warm-up time. - Maximum warm-up time or maximum preparation period from power off to full power condition shall be 20 seconds at ambient temperature of + 10°F. (See 4.7.7)

3.11 Voltage transient. - Under a current surge in the range of 150 to 160 amperes, the instaneous output voltage, as measured across the output, shall not drop below 7.0 volts. Upon recovery of the regulating system from this surge load, the "overshoot" in instantaneous output voltage shall not exceed 5 volts above 27.5 volts dc. The total time of the voltage transient (to reach 17.33 vdc (63 percent of 27.5 vdc)) shall not exceed 200 milliseconds. (See 4.7.8)

3.12 DC efficiency (See 4.7.9). - At ambient temperature of + 75°F, the dc efficiency of the charger shall not be less than the following:

Percent of load	Efficiency (percent)
25	50
50	56
100	57

3.13 Service conditions. - The equipment shall meet the following service conditions:

* 3.13.1 Operation (general). - Continuous use for a period of 5 years, under the worldwide environmental conditions specified in 3.13.2.1 through 3.13.2.6 with a duty cycle of 8 hours on and 4 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement and 6.8. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

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* 3.13.2 Service conditions operation requirements. - Unless otherwise specifed, prior to, during, and after testing as required for each of the following service conditions, the equipment shall conform to the operational test of 4.7.10.

3.13.2.1 Low temperature. - See 4.8.1

3.13.2.2 High temperature. - See 4.8.2

3.13.2.3 Humidity (moisture-resistance). - See 4.8.3

3.13.2.4 Altitude (elevation). - See 4.8.4

3.13.2.5 Orientation. - See 4.8.5

3.13.2.6 Bounce (vibration). - When tested in accordance with 4.8.6, the amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 hertz. Upon completion of the bounce test, the equipment shall show no physical damage except minor surface abrasion, and shall meet the operational requirements of 4.7.10.

* 3.14 Enclosure. - The equipment enclosure shall be of dripproof construction in accordance with MIL-STD-108. After completion of the test of 4.9, the equipment shall meet the operational test of 4.7.10 and there shall be no water in any pocket or bottom of the equipment.

* 3.15 Electromagnetic Interference Characteristics. - The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461, Table A-I-MIL-E-55301, Equipment Class CE. Test and test methods shall be as specified in 4.10.

* 3.16 Overall treatment (moisture and fungus proofing treatment (MFT)). - Moisture and fungus proofing treatment shall be in accordance with MIL-T-152. The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.11. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened surfaces.

* 3.17 Final operational check. - When subjected to the test specified in 4.7.10, the equipment shall meet the requirements of 3.9.2. (See 6.6)

* 3.18 Burn-In/Debugging and bounce preconditioning. - The equipment shall meet the requirements herein, without subsequent processing, after subjection to the burn-in period and bounce preconditioning specified in 4.6. (See 4.5)

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* 3.19 Interchangeability. - Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.12)

3.20 Technical literature, tools, and running spare parts. - Technical literature, tools, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on order. (See 6.2(g) and 6.2(h))

* 3.21 Workmanship. - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In Specification MIL-P-11268:

(1) General requirements for plastic material and plastic pqrts.

(2) Wiring and cabling, including: Slack, protection, clearance, splicing, connections in general, grounding, shielding on wire and cable.

(3) Flux and cleaning agents for solder.

(4) Soldering.

(5) Cleaning.

(6) Riveting.

(7) General requirements for securing of parts.

(8) Welding.

(9) Tropicalization.

(10) Mechanical operation of controls.

(11) Self-tapping screws.

(b) In this Specification:

(1) Marking.

(2) Finish.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.14)

* 4.3 Preproduction inspection. - This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and environmental/design (See tables II, III and I respectively). The inspection requirements of 4.4, group A, and group B shall be performed on all samples prior to environmental/design inspection. Environmental/design inspection shall be performed on the samples specified and in the sequence shown in table I.

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Table I Environmental/design inspection

Inspection	Req. Par.	Insp. Par.
SAMPLE NO. 1		
Low temperature	3.13.2.1	4.8.1
High temperature	3.13.2.2	4.8.2
Warm-up time	3.10	4.7.7
Enclosure	3.14	4.9
Humidity (moisture resistance)	3.13.2.3	4.8.3
SAMPLE NO. 2		
Altitude (elevation)	3.13.2.4	4.8.4
Orientation	3.13.2.5	4.8.5
Electromagnetic interference	3.15	4.10
SAMPLE NO. 3		
Bounce (vibration)	3.13.2.6	4.8.6
Fungus	3.16	4.11

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery.

Item	Where required
Finish	3.5
Marking	3.6

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* 4.5 Quality conformance inspection of equipment before preparation for delivery. - Each unit of each lot of equipment shall be subjected to burn-in/debugging and bounce preconditioning (See 3.18) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test Equipment for Government verification inspection shall be made available by the contractor.

* 4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table II. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.7.10) shall be last. (See 6.6)

Table II - Group A inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and mechanical	3.21	4.13	1.5 DPHU*	6.5 DPHU
Electrical				
Input voltage	3.8	4.7.1)		
Output voltage	3.9.1	4.7.2)	1.5 DPHU	**
Regulation	3.9.1.1	4.7.3)	for the	
Ripple	3.9.1.2	4.7.4)	group	
Efficiency	3.12	4.7.9)		
Output current				
Normal position	3.9.2.1	4.7.5.1)		
Charge position	3.9.2.2	4.7.5.2)		
Operational	3.17	4.7.10	1.0%	**

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* Defects Per Hundred Units.
** All electrical and operational defects are considered major.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to table III and to the procedures for small-sample inspection of MIL-STD-105, using the special inspection levels. The inspection level for this inspection shall be S-4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met Group A inspection.

Table III - Group B inspection

Inspection	Req. Para.	Insp. Para.	AQL
Interchangeability	3.19	4.12	6.5%
Voltage transient	3.11	4.7.8	6.5%
Overload and open circuit protection	3.9.3	4.7.6	6.5%

* 4.5.3 Group C inspection. - This inspection shall consist of the tests listed in Table IV and shall be performed on sample units that have been subjected to and met Groups A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

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Table IV - Group C Inspection

Inspection	Req. Par.	Insp. Par.
Group I		
Orientation	3.13.2.5	4.8.5
Warm-up time	3.10	4.7.7
Group II A		
Low temperature	3.13.2.1	4.8.1
High temperature	3.13.2.2	4.8.2
Humidity (moisture resistance)	3.13.2.3	4.8.3
Group II B		
Altitude (elevation)	3.13.2.4	4.8.4
Bounce (vibration)	3.13.2.6	4.8.6
Group III		
Electromagnetic interference	3.15	4.10
Enclosure	3.14	4.9
Fungus	3.16	4.11

4.5.3.1 Sampling for Group C inspection. -

* 4.5.3.1.1 Group I - Table IV. - One sample shall be selected at random, from each 100 equipments produced and subjected to each inspection specified. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

* 4.5.3.1.2 Group II A and Group II B - Table IV. - One sample shall be selected at random for each group, from each 200 equipments produced and subjected to each inspection specified, in the order shown. The first sample shall be selected at the start of the contract from the first quality conformance inspection production lot. (See 4.5.3)

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4.5.3.1.3 Group III - Table IV. - This inspection shall be required only when preproduction samples are not specified by contract or purchase order. When required, one sample shall be selected at random, at the start of the contract, from the first quality conformance inspection production lot (see 4.5.3) and subjected to each inspection specified. No subsequent inspection for the tests in this group will be required.

4.5.3.2 Noncompliance. - The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable, and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and group A and B inspection may be reinstated; however, final acceptance and shipment will be withheld until the group C inspection results have shown the corrective action was effective. (See 6.5)

4.5.4 Reinspection of conforming group B and group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.

* 4.6 Burn-in/Debugging and bounce preconditioning. - Each equipment shall be subjected to the conditioning period specified in drawing SC-A-678002. After completion of the burn-in/debugging period, the unit shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is possible to insert a 1/32 inch thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection. (See 3.18 and 4.5)

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4.7 Electrical tests. -

* 4.7.1 Input voltage. - Charger, Battery, PP-1451( )/G shall be operated from single phase, ac,variable power source, to determine compliance with the requirement specified in 3.8. Input test voltages and frequencies shall be as specified in the applicable test paragraph.

4.7.2 Output voltage. - With Charger, Battery, PP-1451( )/G connected to the power source specified in paragraph 4.7.1, the output voltage shall be measured with the selector switch in both the NORMAL and CHARGE positions to determine compliance with the requirement specified in 3.9.1. Input test voltages and frequencies shall be as follows:

Input voltage (AC)	Input frequency (HZ)
105V	47.5, 60, 380
115V	60
125V	50, 60, 420
210V	47.5, 60, 380
230V	400
250V	50, 60, 420

* 4.7.3 Regulation (see 3.9.1.1). - With the selector switch in the NORMAL position, Charger, Battery, PP-1451( )/G shall be operated from a power source as specified in paragraph 4.7.1. The load shall be resistive and shall be increased from zero to full in three equal steps under each of the conditions specified in Table V. The percent regulation shall be determined by the following formula:

$$\text{Percent regulation} = 100X \frac{\text{DC Voltage (0 load)} - \text{DC voltage (load)}}{\text{DC voltage (load)}}$$

4.7.4 Ripple (see 3.9.1.2). - With the selector switch in the NORMAL position, Charger, Battery, PP-1451( )/G shall be operated from a power source as specified in paragraph 4.7.1. The load shall be resistive and shall be increased from zero to full in three equal steps under each of the conditions specified in Table V. The percent of ripple shall be determined by the following formula:

$$\text{Percent ripple} = 100X \frac{\text{rms ripple voltage}}{\text{total dc output voltage}}$$

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Table V - Test conditions

Input Voltage, ac	Frequency (Hz)	Output voltage, dc
105	47.5	27.5
105	60	55
105	420	27.5
115	47.5	27.5
115	60	55
115	420	27.5
125	47.5	27.5
125	60	55
125	420	27.5
210	47.5	27.5
210	60	55
210	420	27.5
230	47.5	27.5
230	60	55
230	420	27.5
250	47.5	27.5
250	60	55
250	420	27.5

4.7.5 Output current. -

4.7.5.1 Normal position. - With Charger, Battery, PP-1451( )/G connected to single phase, 115 volts ac, 60 Hz power source; the selector switch in the NORMAL position; and connected to a resistive load, the output current shall be measured to determine compliance with the requirement specified in 3.9.2.1. The above procedure shall be repeated using a 230 volt, 400 Hz ac power source.

4.7.5.2 Charge position. - With the selector switch in the CHARGE position and the voltage control at 27.5 volts, the output current shall be measured while decreasing the load resistance to determine compliance with paragraph 3.9.2.2. (Input Test Voltage: 115 vac - 60 Hz, 230 vac - 400 Hz.)

4.7.6 Overload and open circuit protection. -

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4.7.6.1 Overload and open circuit protection (Voltage applied). - The battery charger shall be tested for circuit protection in both the NORMAL and CHARGE operating positions under the following conditions to determine compliance with paragraph 3.9.3. Input voltage for this test shall be 115 vac, 60 Hz.

(a) Direct short of the dc output terminals.

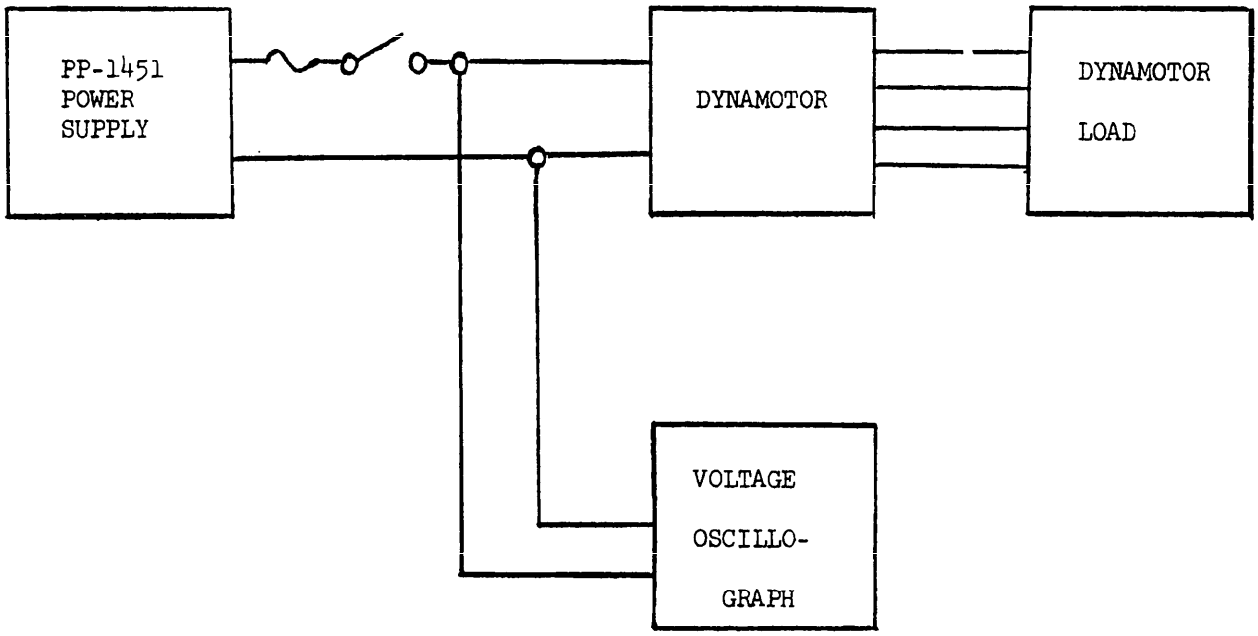
(b) Input circuit "open circuited" while charger is in operation.

4.7.6.2 Open circuit protection (No voltage applied). - The output current contactors of the battery charger shall be checked in both the NORMAL and CHARGE operating positions to determine compliance with paragraph 3.9.3.

* 4.7.7 Warm-up time. - Charger, Battery, PP-1451( )/G shall be placed in a temperature chamber and the temperature lowered to + 10°F (+6 -0)°F. After thermal equilibrium is reached, the time required for equipment to reach full power condition shall be measured to determine compliance with the requirement specified in paragraph 3.10. Input voltage shall be 115 vac, 60 Hz.

* 4.7.8 Voltage transient response. - To determine compliance with the requirement specified in paragraph 3.11, the voltage transient time and excursion shall be measured as shown in Figure 1 using a dynamotor, Windcharger Corporation Model 8S428, or equal. For this test, the output voltage of the equipment under test shall be set at 27.5 volts NORMAL operating position, and the dynamotor adjusted to deliver full load. Input voltage to the battery charger shall be 115 vac, 60 Hz.

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TRANSIENT RESPONSE TIME

FIGURE 1

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4.7.9 Efficiency. - Charger, Battery, PP-1451( )/G shall be operated at 1/4, 1/2, and full load at the following input and output voltages and frequencies to determine compliance with the requirement specified in 3.12.

Input Voltage	Frequency (Hz)	Output voltage
105	50	27.5
105	400	27.5
105	50	55
105	400	55
230	60	27.5
230	60	55
250	60	27.5
250	420	27.5
250	60	55
250	420	55

* 4.7.10 Operational. - Charger, Battery, PP-1451( )/G shall be operated from a single phase, 115 volts ac 60 Hz power source, into a resistive load. The output current shall be measured to determine conformance to 3.17.

* 4.8 Service condition tests. - The following tests shall be performed on Charger, Battery PP-1451( )/G. (See 4.8.1 thru 4.8.6)

* 4.8.1 Low temperature (See 3.13.2.1 and 6.8). - The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be - 80°F for 6 hours and the operating temperature shall be stabilized at - 40°F. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2, 3.9.1.1, and 3.9.1.2.

* 4.8.2 High temperature (See 3.13.2.2 and 6.8). - The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The operating temperature shall be 132°F. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2.

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* 4.8.3 Humidity (moisture-resistance) (See 3.13.2.3 and 6.8). - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet specification performance for the measurements specified in 4.8.3.1. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If the equipment fails to meet specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.8.3.1 Performance. - The voltage output of the equipment shall not be less than 90 percent of the output voltages as measured before the humidity cycle. Four hours after a removal of the equipment from the humidity chamber, the output voltages shall again be measured and shall be not less than 90 percent of the corresponding values obtained prior to the humidity test. The output voltage measurements shall be accomplished in accordance with paragraph 4.7.2 except that the input test voltages and frequencies shall be: (1) 115 vac - 50, 60 and 400 Hz and (2) 230 vac - 60 Hz.

* 4.8.4 Altitude (elevation) (See 3.13.2.4 and 6.8). - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2.

4.8.5 Orientation (See 3.13.2.5 and 6.8). - The equipment shall be inclined for a minimum of 5 minutes in each plane (forward, backward, left, and right) to an angle of 20 (plus 3, minus 1) degrees. Measurements shall be made during inclination in each plane for specification performance as specified in 3.13.2.

* 4.8.6 Bounce (vibration) test. (See 3.13.2.6). - The equipment shall be tested in accordance with Method 514, Procedure XI, of MIL-STD-810. Prior to testing, the equipment shall meet specification performance for the measurements specified in 3.13.2. When practicable, package tester accessories shall be selected from those listed on Drawing SC-A-46439.

* 4.9 Inclosure test. - The battery charger shall be tested in accordance with MIL-STD-108 for dripproof equipment. At the completion of the test, the equipment shall be operated and checked for compliance with the requirements of 3.14.

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* 4.10 Electromagnetic interference characteristics. - Conformance of the test sample in accordance with 3.15 shall be determined by the contractor or an approved testing laboratory. Testing shall be in accordance with MIL-STD-462. The test item shall be operated using a 115 vac, 60 Hz input.

* 4.11 Fungus test (See 3.16). - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation of the equipment is not required.

4.12 Interchangeability. - The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirement of 3.19. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.13 Visual and mechanical inspection. - Equipment shall be examined for the defects listed in MIL-STD-252.

4.14 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

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4.15 Rough handling test (preparation for delivery). - When the rough handling test in accordance with MIL-P-116 is required by contract (see 6.2(d)(2)), the following functional test shall be conducted to determine freedom from operational malfunction caused by the rough handling:

Insp. Para.	Measurement	Input voltage
4.7.10	Operational test	115 vac, 60 Hz

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be Level A or C, as specified. (See 6.2(b).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Battery Charger PP-1451( )/G and the applicable spare parts shall be cleaned in accordance with MIL-P-116, Process C-1.

5.1.1.2 Drying. - Battery Charger PP-1451( )/G and the applicable spare parts shall be dried in accordance with one or more of the procedures listed in MIL-P-116. The procedure used shall not be injurious to the item.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. -

5.1.1.4.1 Technical literature. - Package the technical literature method IC-1 in accordance with MIL-P-116.

5.1.1.4.2 Spare parts. - Package the spare parts individually method III in accordance with MIL-P-116. Place the packaged spare parts within a close fitting fiberboard box conforming to PPP-B-636, class domestic, style optional, normal requirements. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.3 Battery Charger PP-1451( )/G. - Package each Battery Charger PP-1451( )/G individually method 1A-16 in accordance with MIL-P-116 as follows: fabricate a nailed wood box conforming to PPP-B-621, class 2, style 7 to include skids. Drill four (4) holes through the skid base to correspond with the holes in the legs of the battery charger. The holes shall be arranged so that the battery charger is centered on the skid base. The box shall be sized to maintain a two (2) inch clearance between the battery charger and the sides, ends and top of the box. Insert the bolts through the holes provided in the skid base so that the bolt heads are recessed in the skids.

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The diameter of the bolts shall be compatible with the diameter of the holes in the legs of the battery charger. Place a sheet of fiberboard over the bolts conforming to PPP-F-320, class weather-resistant, V3c, sized to fit the skid base. Fabricate a barrier to completely inclose the battery charger with four gasketed holes to correspond with the holes in the skid base. The barrier material shall conform to MIL-B-131, Class 1 and the gaskets shall conform to MIL-G-12803. Place the barrier over the bolts. Apply the required amount of adhesive around each gasketed hole in the barrier. The adhesive shall conform to MIL-A-5092 type III. Fabricate a wrap to completely cushion the battery charger with four holes to correspond with the holes in the skid base. The material used for the wrap shall conform to PPP-P-291, type III, style 2. Place the wrap over the bolts. Place the battery charger on the skid base so that the mounting holes fit over the bolts. Place a lock-washer over each bolt. Place and secure a nut over each bolt. Upset the bolt threads protruding over the nut. Place the spare parts as packaged in paragraph 5.1.1.4.2 on top of the battery charger and secure in place with PPP-T-60 tape, two (2) inches wide. Complete the wrap by placing each end over the unit. Fold the excess material and secure with pressure sensitive tape. Heat seal the barrier leaving a small opening to evacuate the air. Evacuate the air and complete the seal. Place the technical literature package as specified in paragraph 5.1.1.4.1 on top of the battery charger and secure with tape conforming to PPP-T-76 leaving the package markings exposed. Place the hood of the box over the battery charger. Assembly closure and steel strapping shall be in accordance with the appendix of the box specification.

5.1.2 Level C. - Each Battery Charger PP-1451( )/G shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B, or C as specified. (See 6.2(b)).

5.2.1 Level A. - Not required.

* 5.2.2 Level B. - Not required.

5.2.3 Level C. - Each Battery Charger PP-1451( )/G shall be packed in shipping containers in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

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5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Charger, Battery, PP-1451( )/G is intended to be used as follows:

(a) Charge 24-volt or 48-volt batteries. When operating as a battery charger, automatic control circuitry is included which limits the maximum output current to 80 amperes.

(b) Power supply for equipment requiring the output characteristics of the supply, such as Radio Set AN/GRC-19. When operating as a power source for the AN/GRC-19, it will deliver a current surge in the range of 150 to 160 amperes to accommodate the surge currents resulting from the starting of dynamotors in the radio set.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Level of packaging and level of packing required for shipment. (Level A, level B, or level C).

(c) The specific paragraphs of Section 5 which are applicable to this procurement.

(d) Preproduction inspection:

(1) Three sample units of each item cited in the section 1 are generally required so that lengthy and potentially damaging environmental tests can be performed on several sample units. (See 3.2)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon. (See 4.15).

(e) Marking and shipping of samples.

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(f) Place of final inspection.

(g) Technical literature required. (See 3.20)

(h) Quantity of tools and running spare parts required. (See 3.20)

(i) Submission of the statement of treatment covering tropicalization of material as soon as possible after award of contract. This statement should be submitted to the contracting officer. (See 3.16)

6.3 Nomenclature. - The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: PP-1451(W)/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Color. - The color chip furnished by the procuring agency (see 3.5) will be color chip No. X-24087, and may be obtained upon request to Commanding General, Coating and Chemical Laboratory, Aberdeen Proving Grounds, Maryland.

6.5 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Location of operational inspection. - It is desirable that the operational inspection (4.7.10) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.7 Verification inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

* 6.8 Qualitative material requirements. - The following service conditions were considered in the design of the equipment and are given here as a guide for the contractor where he is required to make a selection of parts, materials, processes, constructive methods, etc.

6.8.1 Temperature

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(a) Operating: Ambient temperature in the range of + 132°F to - 40°F. (The 132°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Nonoperating: Exposure in the range of + 160°F to - 80°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

6.8.2 Relative humidity. - Exposure to 97 percent relative humidity for 20 hours; and exposure at 100 percent relative humidity, the condensation, for 4 hours for each 24 hour period.

6.8.3 Altitude (elevation). -

(a) Operating: Up to 10,000 feet above sea level.

(b) Nonoperating: Up to 50,000 feet above sea level.

6.8.4 Orientation. -

(a) Operating. - Any orientation up to 20 degrees from normal operating position (that is: forward, backward, left, or right.)

(b) Nonoperating. - Storage in any position for a period of two years.

NOTE: The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-55211B(EL) Charger, Battery PP-1451()/G

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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